

THE SHEIL–SMALL COVERING PROBLEM FOR SELF-INVERSE POLYNOMIALS IN DEGREES TWO AND THREE

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ABSTRACT. Let $P(z) = \sum_{k=0}^n a_k z^k$ be self-inversive in the standard sense that its zeros are invariant under $\zeta \mapsto 1/\bar{\zeta}$, and put $A = \max_k |a_k|$. A problem of Sheil–Small asks whether $P(\mathbb{D})$ always contains an open disk of radius A . We prove the assertion for $1 \leq n \leq 3$. We also prove it in degrees four and five unless all zeros are unimodular, no endpoint coefficient realizes A , and $2|a_n| < A$; those two puncture strata are left unresolved here. Finally, the quadratic result gives an infinite even-degree subclass supported on $\{0, n/2, n\}$. The proof is analytic and exact. Independent rational Laurent-polynomial verification checks the Schur identities used in the reductions; no proof assistant is used.

1. INTRODUCTION

Write $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$ and $\mathbb{T} = \partial\mathbb{D}$. Problem 4.24 in Hayman and Lingham’s collection asks whether the image of $\mathbb{D}$ under a self-inversive polynomial contains a disk whose radius is the largest coefficient modulus [3]. Their source table traces the problem, attributed to T. Sheil–Small, to the collection edited after the 1973 Canterbury symposium [1]. We use the standard definition of self-inversive: the zero multiset is invariant under $\zeta \mapsto 1/\bar{\zeta}$, equivalently

$$(1) \quad a_k = \omega \overline{a_{n-k}} \quad (0 \leq k \leq n), \quad |\omega| = 1.$$

The coefficient identity is the convention used in the modern literature, for example [4]; its equivalence with zero-multiset invariance is elementary for exact degree and nonzero constant term. A textual caveat is necessary: the accessible source of the Hayman–Lingham edition literally prints $1/\zeta$ in Problem 4.24, without conjugation. No erratum was located. The results below concern the standard conjugate-inversion formulation in (1).

For a domain $\Omega \subset \mathbb{C}$, let

$$\rho(\Omega) = \sup_{c \in \Omega} \text{dist}(c, \mathbb{C} \setminus \Omega).$$

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The question is whether

$$(2) \quad \rho(P(\mathbb{D})) \geq A(P), \quad A(P) = \max_{0 \leq k \leq n} |a_k|.$$

Solyanik proved a binomially weighted coefficient covering bound [5]; for a broader survey of polynomial covering theorems, see Dubinin [2]. Applying Solyanik's Corollary 1 to $P - P(0)$, an endpoint coefficient of modulus $A(P)$ already yields the desired disk. The same fact follows from the elementary root-product argument in Lemma 2.1 below. We therefore separate that known branch from the low-degree reductions.

Our main result is the following. All degrees are exact and all zeros are counted with multiplicity.

Theorem 1.1. *Let P satisfy (1).*

- (1) *If $1 \leq n \leq 3$, then (2) holds.*
- (2) *If $n \in \{4, 5\}$, then (2) holds provided at least one of the following conditions holds:*
 - (a) $|a_0| = |a_n| = A(P)$;
 - (b) *some zero of P is off $\mathbb{T}$;*
 - (c) *all zeros of P are on $\mathbb{T}$ and $2|a_n| \geq A(P)$.*
- (3) *If $n = 2m$ and P is supported on $\{0, m, 2m\}$, then (2) holds.*

The two unresolved low-degree strata are precisely

$$(3) \quad n \in \{4, 5\}, \quad \text{all zeros lie on } \mathbb{T}, \quad |a_n| < A(P), \quad 2|a_n| < A(P).$$

Thus Theorem 1.1 is a partial theorem for the original problem, not a solution in all degrees and not a complete decision in degrees four and five. A claim-specific search through August 29, 2026 did not locate the degree-two, degree-three, or non-endpoint partial results in the literature; the negative search statement is necessarily bounded by the databases queried.

2. BOUNDARY AND SCHUR LEMMAS

Multiplication of P by a nonzero scalar and precomposition by a rotation preserve (2). We may therefore choose the Hermitian convention

$$(4) \quad a_k = \overline{a_{n-k}}.$$

For $c \in \mathbb{C}$ and $r > 0$, write $B(c, r) = \{w : |w - c| < r\}$.

Lemma 2.1 (endpoint disk). *If $a_n \neq 0$, then $B(a_0, |a_n|) \subset P(\mathbb{D})$.*

Proof. For $|w - a_0| < |a_n|$, the product of the moduli of the roots of $P - w$ is $|a_0 - w|/|a_n| < 1$. At least one root lies in $\mathbb{D}$. $\square$

Lemma 2.2 (boundary preimages). *Put $\Omega = P(\mathbb{D})$. If $w \in \partial\Omega$, every zero of $P - w$ has modulus at least one and at least one has modulus one. Consequently $w = P(z)$ for some $z \in \mathbb{T}$, and every zero of*

$$\frac{P(x) - P(z)}{x - z}$$

has modulus at least one.

Proof. An interior preimage would make w interior to Ω by the open mapping theorem. If every root had modulus strictly greater than one, the finite root multiset would have positive separation from $\mathbb{T}$. Root continuity, or equivalently Rouché's theorem on $\mathbb{T}$, preserves that separation for nearby values, making w interior to the complement. Both alternatives contradict $w \in \partial\Omega$. Division by a unimodular root gives the last assertion. $\square$

In particular, if $0 \in \partial\Omega$, then every zero of a self-inversive P lies on $\mathbb{T}$: an outside zero would have an inside reciprocal-conjugate mate. Conversely, if every zero lies on $\mathbb{T}$, then $0 \notin \Omega$ and radial approach to a zero shows $0 \in \partial\Omega$.

For $H(y) = h_0 + h_1y + \cdots + h_my^m$, define

$$H^\#(y) = \sum_{j=0}^m \overline{h_{m-j}} y^j, \quad \mathcal{S}H(y) = \frac{\overline{h_m}H(y) - h_0H^\#(y)}{y}.$$

Lemma 2.3 (Schur reduction). *Suppose all zeros of H lie in $\overline{\mathbb{D}}$. If $|h_0| < |h_m|$, then $\mathcal{S}H$ has degree $m-1$ and all its zeros lie in $\overline{\mathbb{D}}$. If $|h_0| = |h_m|$, all zeros of H are unimodular and $\mathcal{S}H \equiv 0$.*

Proof. The product of the root moduli is $|h_0/h_m|$. If H has no zero on $\mathbb{T}$, then on $\mathbb{T}$ one has $|H^\#| = |H|$ and $|h_0H^\#| < |\overline{h_m}H|$. Rouché's theorem gives the numerator the same m disk zeros as H ; removing its zero at the origin leaves $m-1$. If H has boundary zeros, move those roots radially into $\mathbb{D}$ while keeping the leading coefficient fixed. The strict result applies to the approximants, and coefficient convergence and continuity of roots put all zeros of the limiting transform in $\overline{\mathbb{D}}$. Its leading coefficient is $|h_m|^2 - |h_0|^2 \neq 0$, so its degree is still $m-1$. In the equality case the root-product formula forces every root modulus to be one. Then coefficient reversal is proportional to H and the displayed numerator vanishes identically. $\square$

3. DEGREES TWO AND THREE

Degree one follows immediately from Lemma 2.1.

Proposition 3.1. *Every self-inversive quadratic satisfies (2).*

Proof. The allowed phase and rotation normalizations give

$$P(z) = r + bz + rz^2, \quad r > 0, \quad b \in \mathbb{R}.$$

Replacing z by $-z$, assume $b \geq 0$. If $r \geq b$, apply Lemma 2.1. If $b \geq r$, then for $z = e^{it}$,

$$z^{-1}\{P(z) - 2r\} = b + r(z - z^{-1}) = b + 2ir \sin t,$$

so $|P(z) - 2r| \geq b$ on $\mathbb{T}$. At the center, the roots of $rz^2 + bz - r$ are real with product -1 ; for $b > 0$ exactly one lies in $\mathbb{D}$. Rouché's theorem preserves that interior root throughout $B(2r, b)$. The case $b = 0$ was already in the endpoint branch. Thus a disk of radius $\max(r, b) = A(P)$ is contained in the image. $\square$

Proposition 3.2. *Every self-inversive cubic satisfies (2).*

Proof. The endpoint-maximal case is Lemma 2.1. Otherwise rotations, phase, and scaling give

$$(5) \quad P(z) = \bar{d} + z + z^2 + dz^3, \quad |d| \leq 1, \quad A(P) = 1.$$

Let $w = P(z) \neq 0$ be a boundary value with $z \in \mathbb{T}$. The reciprocal of the quotient in Lemma 2.2 is

$$H_z(y) = d + (dz + 1)y + (dz^2 + z + 1)y^2.$$

Its zeros lie in $\overline{\mathbb{D}}$. Its nominal leading coefficient cannot vanish, since that would make 0 a root of the unreversed quotient. Direct expansion gives

$$(6) \quad \mathcal{S}H_z(y) = \frac{P(z)}{z^2} \{1 + (1 + z)y\}.$$

The equality branch of Lemma 2.3 is impossible because the right side is not the zero polynomial. Stability of the remaining linear factor yields $|1 + z| \geq 1$.

Choose $z = e^{it}$ with $t \in [-\pi, \pi]$, and put $x = \cos(t/2) \in [1/2, 1]$ and

$$(7) \quad C = \frac{1}{2} + 2\bar{d}.$$

The d -terms in $z^{-3/2}(P(z) - C)$ are purely imaginary. Hence

$$(8) \quad |P(z) - C| \geq 2 \cos(t/2) - \frac{1}{2} \cos(3t/2) = \frac{7}{2}x - 2x^3 \geq \frac{3}{2}.$$

The last polynomial is concave on $[1/2, 1]$ and has value $3/2$ at both endpoints.

It remains to control the possible boundary value zero. If zero is a boundary value, all roots of P are unimodular. Gauss–Lucas puts the roots of $P'(z) = 1 + 2z + 3dz^2$ in $\mathbb{D}$. Its root product and one Schur step give

$$(9) \quad |d| \geq \frac{1}{3}, \quad |6\bar{d} - 2| \leq 9|d|^2 - 1.$$

For $r = |d|$, squaring the second inequality yields

$$\operatorname{Re} d \geq \frac{1 + 18r^2 - 27r^4}{8}.$$

Consequently

$$|C|^2 \geq \frac{1}{2} + \frac{17}{2}r^2 - \frac{27}{4}r^4 \geq \frac{49}{36}.$$

The last expression is concave as a function of $r^2 \in [1/9, 1]$, so its minimum is at an endpoint.

Finally set

$$P_s(z) = s\bar{d} + z + z^2 + sdz^3, \quad C_s = \frac{1}{2} + 2s\bar{d}, \quad 0 \leq s \leq 1.$$

At $s = 0$, $P_0(z) = C_0$ has the interior root $(\sqrt{3} - 1)/2$. This root persists for small positive s , while the new third root enters from infinity and is outside the disk. For $s > 0$, the number of roots of $P_s - C_s$ in $\mathbb{D}$ is locally constant

away from unit-circle roots. If membership failed at $s = 1$, there would be a parameter s_* at which the last interior root is lost; then C_{s_*} would be a boundary value. If it were nonzero, (8) would give $0 \geq 3/2$; if it were zero, the preceding all-unimodular estimate would give $|C_{s_*}| \geq 7/6$. Thus $C = C_1 \in P(\mathbb{D})$.

Every nonzero boundary point is at distance at least $3/2$ from C , and a zero boundary point is at distance at least $7/6$. A segment from C to an omitted point of $B(C, 1)$ would meet the boundary inside that disk, a contradiction. Hence $B(C, 1) \subset P(\mathbb{D})$. $\square$

Remark 3.3. The non-endpoint proof actually gives $B(C, 7/6)$ in the all-unimodular cubic case and $B(C, 3/2)$ otherwise. Equality $|C| = 7/6$ occurs for $P = (1 + z)^3/3$.

4. PARTIAL RESULTS IN DEGREES FOUR AND FIVE

Proposition 4.1. *Let P be a self-inversive quartic. If an endpoint coefficient realizes $A(P)$, if some zero is off $\mathbb{T}$, or if all zeros are unimodular and $2|a_4| \geq A(P)$, then (2) holds.*

Proof. Only the non-endpoint case needs proof. In the Hermitian convention write

$$(10) \quad P(z) = \bar{d} + \bar{b}z + cz^2 + bz^3 + dz^4, \quad c \in \mathbb{R}, \quad A = \max(|b|, |c|) > |d|.$$

At a nonzero boundary value $P(z)$, $z \in \mathbb{T}$, the reciprocal boundary quotient has coefficient list

$$[d, dz + b, dz^2 + bz + c, dz^3 + bz^2 + cz + \bar{b}].$$

Its first Schur reduction is the nonzero scalar $P(z)/z^3$ times

$$K_z(y) = b + (bz + c)y + (bz^2 + cz + \bar{b})y^2.$$

Thus K_z is stable. Put $u = bz$ and $S = c + u + \bar{u} \in \mathbb{R}$. Its leading coefficient is zS , so $S \neq 0$ and root products give $|S| \geq |b|$. The constant and leading coefficients in the next Schur step, up to unit factors, are

$$cS + u^2, \quad S^2 - |b|^2.$$

The strict and equal-product branches therefore both give

$$|cS + u^2| \leq S^2 - |b|^2.$$

The reverse triangle inequality implies $|S| \geq |c|$, hence $|S| \geq A$. On $\mathbb{T}$,

$$(11) \quad z^{-2}\{P(z) - 2\bar{d}\} = S + dz^2 - \bar{d}z^{-2},$$

and the final two terms are purely imaginary. Every nonzero boundary point is therefore at distance at least A from $2\bar{d}$.

The homotopy obtained by replacing d with sd and the center with $2s\bar{d}$ starts with the interior preimage $z = 0$. Rouché on a small circle about zero preserves at least one nearby interior root for small positive s . If membership failed at $s = 1$, the last interior root would be lost at a nonzero boundary

value, which is impossible by (11). Thus $2\bar{d} \in P(\mathbb{D})$. If a zero is off $\mathbb{T}$, reciprocal-conjugate symmetry supplies an interior zero; hence 0 is not a boundary value and the full disk $B(2\bar{d}, A)$ is contained in the image. If all zeros are unimodular, the same argument gives only

$$B(2\bar{d}, A) \setminus \{0\} \subset P(\mathbb{D}).$$

When $2|d| \geq A$, zero does not belong to the open disk, completing the stated case. $\square$

Proposition 4.2. *A self-inversive quintic satisfies (2) if an endpoint is maximal, if a zero is off $\mathbb{T}$, or if all zeros are unimodular and $2|a_5| \geq A(P)$.*

Proof. Again treat only the non-endpoint case and write

$$P(z) = \bar{d} + \bar{b}z + \bar{c}z^2 + cz^3 + bz^4 + dz^5, \quad A = \max(|b|, |c|) > |d|.$$

At a nonzero boundary value, reversing the boundary quotient and taking the first Schur reduction gives, up to the nonzero factor $P(z)/z^4$,

$$K_z = [b, bz + c, bz^2 + cz + \bar{c}, bz^3 + cz^2 + \bar{c}z + \bar{b}].$$

Choose $z^{1/2}$ and put

$$B = bz^{3/2}, \quad C = cz^{1/2}, \quad S = B + C + \bar{C} + \bar{B} \in \mathbb{R}.$$

Unit changes of variable and scale send K_z to

$$L(y) = B + (B + C)y + (B + C + \bar{C})y^2 + Sy^3.$$

The first transform has exact degree three, so $S \neq 0$. Stability gives $|S| \geq |B|$. The next Schur step has endpoint coefficients $SC + B^2$ and $S^2 - |B|^2$, whence

$$|SC + B^2| \leq S^2 - |B|^2 \implies |S| \geq |C|.$$

Therefore $|S| \geq A$. The exact boundary decomposition

$$z^{-5/2}\{P(z) - 2\bar{d}\} = S + dz^{5/2} - \bar{d}z^{-5/2}$$

places every nonzero boundary value at distance at least A from $2\bar{d}$. The endpoint homotopy, the off-circle zero argument, and the punctured-disk argument are identical to the quartic proof. They give the full disk when some zero is off $\mathbb{T}$, and in the all-unimodular case they give

$$B(2\bar{d}, A) \setminus \{0\} \subset P(\mathbb{D}).$$

This is the required disk when $2|d| \geq A$. $\square$

Remark 4.3 (the puncture gaps). For $n = 4, 5$, the preceding proof controls every nonzero boundary value but does not fill the isolated boundary value zero on (3). For example, $(1 + z)^4/6$ lies in the quartic residual stratum, and zero lies inside the displayed candidate disk. Deleting the puncture is therefore not a valid completion. No assertion about (2) on either residual stratum is made here.

5. AN INFINITE SPARSE SUBCLASS

Corollary 5.1. *Let $n = 2m$. If a self-inversive polynomial of degree n is supported on $\{0, m, 2m\}$, then it satisfies (2).*

Proof. Write $P(z) = Q(z^m)$, where Q is a self-inversive quadratic and $A(P) = A(Q)$. Since z^m maps $\mathbb{D}$ onto $\mathbb{D}$, the images $P(\mathbb{D})$ and $Q(\mathbb{D})$ coincide. Proposition 3.1 applies. $\square$

For every positive degree, an endpoint binomial has the exact image

$$(a_0 + a_n z^n)(\mathbb{D}) = B(a_0, |a_n|).$$

This elementary observation is included only for orientation and is not a novelty claim.

6. EXACT VERIFICATION AND SCOPE

The displayed Schur and Laurent identities were reconstructed in two independent exact implementations. One uses SymPy; the other uses only Python integers, rational numbers, and sparse Laurent monomials. A third fail-closed verifier loads a canonical JSON certificate and recomputes eleven identities from serialized input. All checks pass in normal, optimized, isolated, and optimized-isolated Python modes. Intentional coefficient corruption and injected-failure tests exit nonzero.

These scripts verify algebraic identities, not the analytic use of the open mapping theorem, Rouché’s theorem, Gauss–Lucas, or the root-count homotopies. Those arguments are given above. No floating-point search is used in any proof. Numerical searches through degree twelve found no robust counterexample, but they are diagnostics only and are not evidence for an unproved theorem.

Through August 29, 2026, our searches located no resolution of the unrestricted standard Sheil–Small problem or either stratum in (3); all three are unresolved in the present work. No Lean, Coq, Isabelle, HOL, or other proof assistant was used.

COMPUTATIONAL AND AI DISCLOSURE

OpenAI Codex was used to assist literature searching, symbolic factorization, numerical diagnostics, drafting, and adversarial proof checking. The mathematical claims were separated from all numerical experiments, independently reconstructed from the definitions, and checked by fail-closed exact programs. The author remains responsible for the statements and proofs.

DATA AND CODE AVAILABILITY

The source package accompanying this manuscript contains the formal problem statement, proof and gap ledgers, the serialized exact certificate,

three verification programs, dependency pins, two novelty audits, the citation audit, and the independent referee report. Reproduction commands and SHA-256 bindings are recorded in the package manifest.

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